

CACS101: Computer Fundamentals and Applications

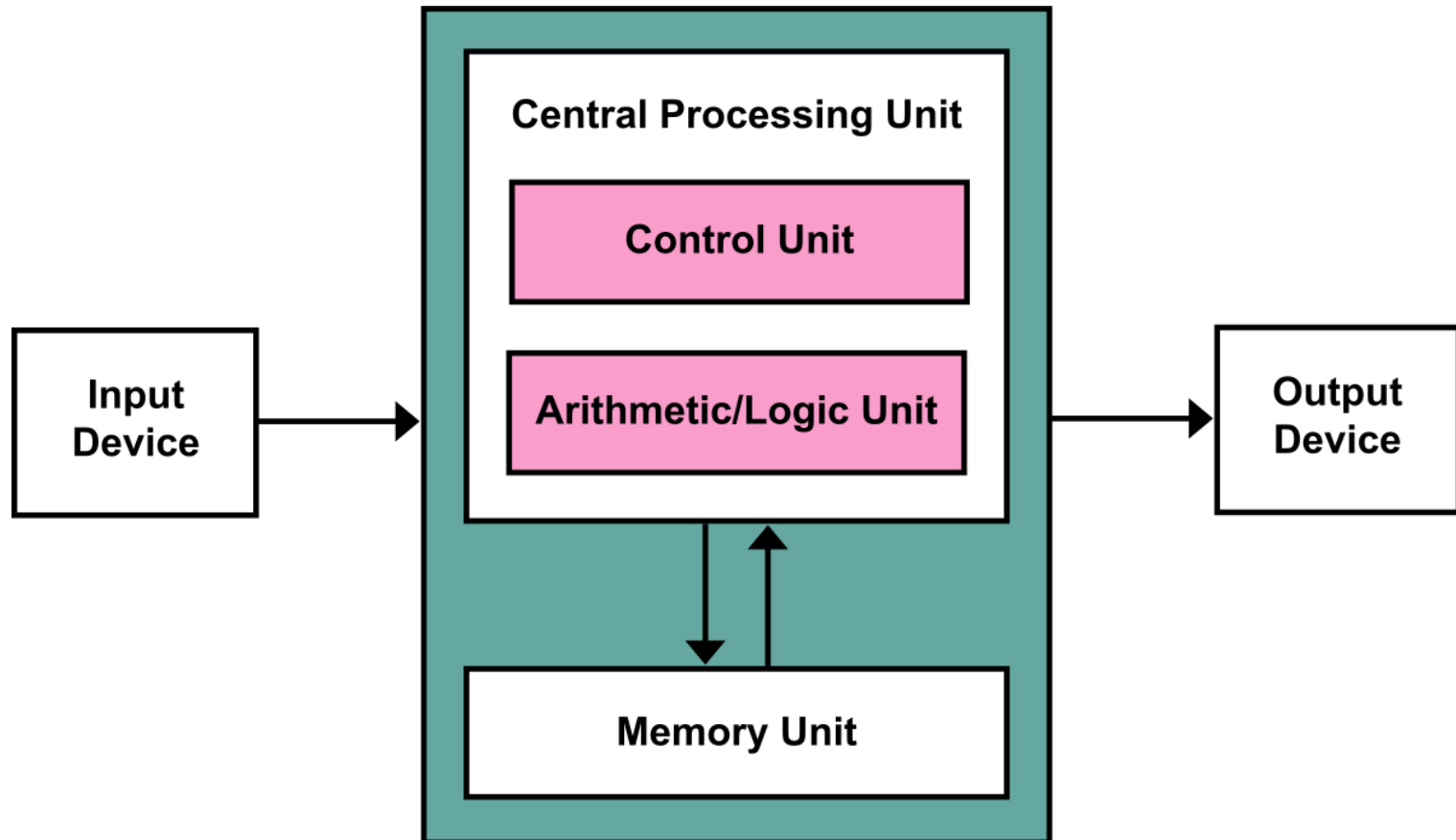
Unit 1 – Introduction to Computer System

PRAJWAL BISTA

Introduction to Computer

- The word "computer" is derived from the Latin word "computare" which means "to calculate".
- Computer is an electronic device which has the ability to retrieve, store, process data.
- The only language supported by computer system is "Binary System" or "Machine Language". The machine language is represented by 0's and 1's.

Computer Architecture



Computer Architecture

- Computer Architecture refers to the definition of the basic attributes of hardware components and their interconnections, in order to achieve certain specified goals in terms of functions and performance.
- The attributes may include the instruction set, data representation, I/O mechanisms, etc.
- The architecture basically defines the logical structure of a computer system.
- It is a specification detailing how a software and hardware technology standards interact to form a computer system.

CISC Architecture

- CISC stands for Complex Instruction Set Computer. CISC chips have a larger set of complex instructions.
 - The CISC machines have complex instructions that need a multiple number of cycles for their execution.
 - Variable length Instruction format[8 or 16 or 32 or 64 bits]
 - Large number of addressing modes.
 - Cost is high.
 - More powerful.
 - Several Cycle instructions.
 - Emphasis on hardware.
 - Manipulation directly in memory.
- MULT 2:2, 3:3

RISC Architecture

- RISC stands for Reduced Instruction Set Computer.
- The RISC machines have simpler instructions that need a single cycle for their execution.
- Fixed length Instruction format[32]
- Few number of addressing modes.
- Cost is low.
- Less powerful.
- Single Cycle instruction.

LOAD A, 2:2

LOAD B, 3:3

PROD A, B

STORE 2:2, A

Characteristics of Computer

1. Speed

A computer works with much higher speed and accuracy compared to humans while performing mathematical calculations. Computers can process millions of instructions per second. The time taken by computers for their operations is microseconds and nanoseconds.

2. Accuracy

Computers perform calculations with 100% accuracy. Errors may occur due to data inconsistency or inaccuracy.

Contd.

3. Diligence

A computer can perform millions of tasks or calculations with the same consistency and accuracy. It doesn't feel any fatigue or lack of concentration. Its memory also makes it superior to that of human beings.

4. Versatility

Versatility refers to the capability of a computer to perform different kinds of works with same accuracy and efficiency.

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5. Reliability

A computer is reliable as it gives consistent result for similar set of data i.e., if we give same set of input any number of times, we will get the same result.

6. Automation

Computer performs all the tasks automatically i.e. it performs tasks without manual intervention.

7. Memory

A computer has built-in memory called primary memory where it stores data. Secondary storage are removable devices such as CDs, pen drives, etc., which are also used to store data.

Applications of Computer

- Home

Computers are used at homes for several purposes like online bill payment, watching movies or shows at home, home tutoring, social media access, playing games, internet access, etc. They provide communication through electronic mail. They help to avail work from home facility for corporate employees. Computers help the student community to avail online educational support.

Contd.

- Medical Field

Computers are used in hospitals to maintain a database of patients' history, diagnosis, X-rays, live monitoring of patients, etc. Surgeons nowadays use robotic surgical devices to perform delicate operations, and conduct surgeries remotely. Virtual reality technologies are also used for training purposes. It also helps to monitor the fetus inside the mother's womb. ECG's, radiotherapy wasn't possible without computers.

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- **Entertainment**

Different types of entertainment fields such as multimedia, film making with animation, graphics, audio and visual design are done with the help of computer.

- **Military**

Computers are the main tools which help in developing missiles and other equipment in the defense system. Designing and the maintenance are possible only through computers. Computer builds the links between the soldiers and commanders through the satellite. Construction of weapons and controlling their function is not possible without the aid of computers. The list of the criminals and the records of the cops are maintained regularly in the system.

Contd.

- Industry

Computers are used to perform several tasks in industries like managing inventory, designing purpose, creating virtual sample products, interior designing, video conferencing, etc. Online marketing has seen a great revolution in its ability to sell various products to inaccessible corners like interior or rural areas. Stock markets have seen phenomenal participation from different levels of people through the use of computers.

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- Education

Computers are used in education sector through online classes, online examinations, referring e-books, online tutoring, etc.

- Government

In government sectors, computers are used in data processing, maintaining a database of citizens and supporting a paperless environment. The country's defense organizations have greatly benefitted from computers in their use for missile development, satellites, rocket launches, etc.

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- Banking

In the banking sector, computers are used to store details of customers and conduct transactions, such as withdrawal and deposit of money through ATMs. Banks have reduced manual errors and expenses to a great extent through extensive use of computers.

- Business

The main objective of business is transaction processing, which involves transactions with suppliers, employees or customers. Computers can make these transactions easy and accurate. People can analyze investments, sales, expenses, markets and other aspects of business using computers.

Contd.

- Science and Engineering

Computers with high performance are used to stimulate dynamic process in Science and Engineering. Supercomputers have numerous applications in area of Research and Development (R&D). Topographic images can be created through computers. Scientists use computers to plot and analyze data to have a better understanding of earthquakes.

Classification of Computer

- **On the basis of Size and Speed**
- **On the basis of working principle**

On the basis of Size and Speed

- Super Computer
- Mainframe Computer
- Mini Computer
- Micro Computer

Super Computer

- A Supercomputer is the **fastest** and **powerful** computer for processing data. Its' size and storage capacity is also huge; designed to process vast amounts of data in a short time with high productivity. It can execute hundreds of millions of instructions per second.
- These are specially made to perform **multi-specific tasks**. Therefore, many CPUs work in parallel order on these supercomputers. This function of a Supercomputer is called Multi or parallel processing.
- Examples: IBM Roadburner, IBM Blue Gene, CRAY-XMP-14.

Contd.

- Although both desktop computers and supercomputers are equipped with similar processors, their speed and memory sizes are significantly different. For instance, a desktop computer built in the year 2000 normally has a hard disk data capacity of between 2 and 20 gigabytes and one processor with tens of megabytes of random access memory (RAM)-just enough to perform tasks such as word processing, web browsing, and video gaming. Meanwhile, a supercomputer of the same time period has thousands of processors, hundreds of gigabytes of RAM, and hard drives that allow for hundreds, and sometimes thousands, of gigabytes of storage space.

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- The supercomputer's large number of processors, enormous disk storage, and substantial memory greatly increase the power and speed of the machine. Although desktop computers can perform millions of floating-point operations per second (megaflops), supercomputers can perform at speeds of billions of operations per second (gigaflops) and trillions of operations per second (teraflops).

Applications of Super Computer

- In research and study of energy and nuclear weapons and designing the aircraft, airplanes, and flight simulators.
- Climate research and Weather Forecasting and Prediction of Natural Disasters.
- Spaceship and Satellite Launching.
- Used in scientific research laboratories.
- Used in Chemical and Biological research and for highly calculation complex tasks.

Mainframe Computer

- Mainframe computers are **multi-programming, high-performance** computers, and multi-user, which means they can handle the workload of more than 100 users at a time on the computer.
- The storage capacity of the mainframe is enormous, with a high-speed data process as well. As well as handling hundreds of input and output devices at a time.

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- The mainframe is a highly efficient computer capable of **simultaneously solving complex calculations and continuously for a long time**. These computers have several microprocessors that have the ability to function the data at too high performance and speed.
- Examples: IBM Es000 series, ICL39 Series, CDC 6600.

Applications of Mainframe Computer

- Mainframe computers are mainly used by departmental and commercial organizations like Banks, Companies, Scientific research centers, and railways.
- These computers can work for 24 hours. Hundreds of users can work on these computers simultaneously.

Mini Computer

- Minicomputer is a **digital and multi-user** computer system with the connection of more than one CPU. Thus, many people can work on these computers simultaneously instead of a single person.
- Minicomputers are the medium type of computers that have more functionality power and are expensive than microcomputers. On the other hand, the size, storage, and speed of minicomputers are **large but less than** the mainframe and supercomputers.

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- Minicomputers are made for performing multiple computing tasks at a single point of time, instead of assigning many microcomputers for a single task, which will be time-consuming and expensive.
- In general, a minicomputer is a multi-threaded system (several processes at a time) capable of supporting from one to up to 200 users simultaneously: these computers are currently used to *store large databases, multi-user applications, and the automation industry*.
- Examples: PDP 11 and IBM (8000 Series).

Applications of Minicomputers

- The minicomputers are used as real-time applications in Industries, bookings, and Research Centers. Banks also use minicomputers for preparing payroll for employees' salaries, records, tracking of financial accounts, etc. As well as in the field of Higher Education and Engineering.

Micro Computer

- With the invention of the microprocessors in the year 1970, it became possible to use computers for people personally at a low cost and reasonable price known as Digital **Personal Computer**.
- The design of microcomputers is minimal in size and storage capacity. These computers consist of many parts like *Input and Output devices, Software, operating systems, networks, and Servers* all these need to connect to form a complete Personal Digital Computer.

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- **Desktop Computer:** a personal or micro-mini computer sufficient to fit on a desk.
- **Laptop Computer:** a portable computer complete with an integrated screen and keyboard. It is generally smaller in size than a desktop computer and larger than a notebook computer.
- **Palmtop Computer/Digital Diary /Notebook /PDAs:** a hand-sized computer. Palmtops have no keyboard but the screen serves both as an input and output device.

Applications of Micro Computer

- PC is being widely used in many fields like home, office, data collection, business, education, entertainment, publishing, etc.
- It keeps the details and prepares letters for correspondence in small businesses, creating bills, accounting, word processing, and operation of the filing systems in a large company.

On the basis of working principle

- Analog Computer
- Digital Computer
- Hybrid Computer

Analog Computer

- An analog computer performs tasks using continuous data. Analog computers are used primarily to measure physical units like the voltage, pressure, electric current, temperature.
- Analog computers are mainly used in the fields of science and engineering. Analog computers are less reliable than digital computers.

Contd.

- They are used in the fields of **technology, science, research, engineering**, etc. Because quantities like **voltage, pressure, electric current, temperature** are used more in these areas, these types of computers give only approximate estimates.
- Examples: Analog clock, thermometer, **vehicle's speedometer, barometer**, etc. are examples of analog computing.

Digital Computer

- Computers which work on discrete signals (digital signals)
- It can perform arithmetic operations such as ***addition, subtraction, multiplication, or division and all types of logical(mathematical) operations.***

Contd.

- Digital computers use discrete electrical signals for operation making them the most common form of computers today because of their **versatility, speed, and power**.
- Examples: Personal Desktop Computers, Calculators, Laptops, Smartphones, and Tablets, Digital Watch, etc.

Analog Computer vs Digital Computer

Analog Computer	Digital Computer
Analog computers are less reliable than digital computers.	Digital computers are more reliable than analog computers.
Its performance is comparatively low.	Its performance is very high.
Its speed of processing is not so high.	Its speed of processing is very high.
It provides results with less accuracy as compared to digital computers.	It provides results with higher accuracy as compared to analog computers.
They are usually special purpose devices.	It can be general purpose devices.
Examples includes analog clock and thermometer etc.	Examples includes Digital laptop, digital camera, digital watches etc.

Hybrid Computer

- A hybrid computer is a combined complex computer unit built using both ***analog and digital*** properties and united by a single control system. The purpose of designing hybrid computers is to offer best of both types of computers.
- **Hybrid computers are extremely fast** when driving equations, even when those calculations are incredibly complex.

Contd.

- As both the properties of a computer consist in the form of a single hybrid computer made possible ***to solve too complex calculations or problems***, at the same time, these devices are not only too expensive but also able to solve complex issues a bit fastly.
- The analog computer systems solve the equation process immediately. But it is not necessary that those solutions would be completely accurate.

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- Here, the digital computer system gives 100 % correct solutions, but it takes a bit of time in the calculation. Therefore, Hybrid computers are made to overcome the flaws of both these analog and digital systems.
- Example: **Auto Gasoline pump, Electrocardiogram Machine, Monitoring Machine, etc.**

Contd.

- Hybrid computers are most commonly used in vast industries, research centers, organizations, and manufacturing firms (where many equations need to be solved).
- Also, the solutions and uses of hybrid computers have proved to be much more detailed, accurate, and useful. Hybrid computers are used in scientific calculations, for nations' defense and radar systems as well.

Mobile Computing

- Mobile computing is a technology that enables data, voice, and video to be transmitted wirelessly from a computer or other wireless enabled device without the need for a fixed physical connection in a timely, cost-effective way.
- The concept of Mobile Computing can be divided into three parts:
 - Mobile Communication
 - Mobile Hardware
 - Mobile Software

Mobile Communication

- Mobile Communication specifies a framework that is responsible for the working of mobile computing technology.
- It refers to an infrastructure that ensures **consistent, seamless and reliable communication**; among wireless devices.
- It consists of communication devices such as protocols, bandwidth necessary to facilitate and support the stated services. These devices are responsible for delivering a smooth communication process.
- The wireless networks utilized in communication are Bluetooth, W-LANs, Cellular and satellite communication system.

Types

- **Fixed and Wired:** In Fixed and Wired configuration, the devices are fixed at a position, and they are connected through a physical link to communicate with other devices. *Example: Desktop*
- **Fixed and Wireless:** In Fixed and Wireless configuration, the devices are fixed at a position, and they are connected through a wireless link to make communication with other devices. *Example: Communication Tower, WiFi router, Temporary Setups*

Types

- **Mobile and Wired:** In Mobile and Wired configuration, some devices are wired, and some are mobile. They all together make communication with other devices. *Example: Laptop*
- **Mobile and Wireless:** In Mobile and Wireless configuration, the devices can communicate with each other irrespective of their position. They can also connect to any network without the use of any wired device. *Example: Laptop with WLAN*

Mobile Hardware

- Mobile hardware is a portable computing device with the ability to retrieve and process data.
- Examples of mobile hardware can be smartphones, laptops, portable PCs, tablet PCs, Personal Digital Assistants, etc.
- These devices are inbuilt with a receptor medium that can send and receive signals. These devices are capable of operating in **full-duplex**. It means they can send and receive signals at the same time. They don't have to wait until one device has finished communicating for the other device to initiate communications.

Mobile Software

- Mobile software, an essential component that operates the mobile device is a program that runs on mobile hardware. This is designed to deal capably with the characteristics and requirements of mobile applications.
- This is the **operating system** in mobile devices.
- These operating systems provide features such as touchscreen, cellular connectivity, Bluetooth, Wi-Fi, GPS mobile navigation, camera, video camera, speech recognition, voice recorder, music player, near field communication and sensors.

Advantages of Mobile Computing

- Seamless and reliable communication
- Increased productivity
- Entertainment
- Portability
- Cloud Computing
- Web or internet access
- GPS (Global Positioning System)
- Emergency services (hire an ambulance, police call, etc)

Disadvantages of Mobile Computing

- The obstacle to battery consumption.
- The transmission bandwidth is inefficient.
- Over the whole network, there are link losses.
- Fluctuation in the stability of the network.
- Small screen sizes.

Anatomy of a Digital Computer

- The computer system consists of three main components:
 - Input/Output (I/O) Unit,
 - Central Processing Unit (CPU), and
 - Memory Unit.
- The I/O unit consists of the input unit and the output unit. CPU performs calculations and processing on the input data, to generate the output. The memory unit is used to store the data, the instructions and the output information.

Input/Output Unit

- The user interacts with the computer via the I/O unit. The Input unit accepts data from the user and the Output unit provides the processed data.
- The Input unit converts the data that it accepts from the user, into a form that is understandable by the computer. Similarly, the Output unit provides the output in a form that is understandable by the user. The input is provided to the computer using input devices like keyboard, trackball and mouse. Some of the commonly used output devices are monitor and printer.

Central Processing Unit (CPU)

- CPU **controls, coordinates and supervises** the operations of the computer. It is responsible for processing of the input data.
- CPU consists of Arithmetic Logic Unit (ALU) and Control Unit (CU).
- ALU performs all the arithmetic and logic operations on the input data.
- CU controls the overall operations of the computer i.e. it checks the sequence of execution of instructions, and, controls and coordinates the overall functioning of the units of computer.
- Additionally, CPU also has a set of registers for temporary storage of data, instructions, addresses and intermediate results of calculation.

Memory Unit

- Memory unit stores the data, instructions, intermediate results and output, **temporarily**, during the processing of data. This memory is also called the main memory or primary memory of the computer.
- The **input data** that is **to be processed** is brought into the main memory before processing. The **instructions** required for processing of data and any intermediate results are also stored in the main memory.
- The output is stored in memory **before being transferred** to the output device. CPU can work with the information stored in the main memory.
- Another kind of storage unit is also referred to as the secondary memory of the computer. The data, the programs and the output are stored permanently in the storage unit of the computer. Magnetic disks, optical disks and magnetic tapes are examples of secondary memory.

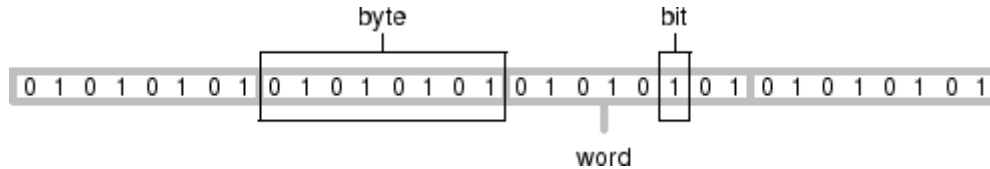
Computer Memory

- Computer memory is the storage space in the computer, where **data is to be processed** and **instructions required for processing** are stored.
- Different types of memories are available in a computer. The cache memory, registers, and RAM are **fast** memories and store the data and instructions **temporarily** during the processing of data and instructions.
- The secondary memory like magnetic disks and optical disks have large storage capacities and store the data and instructions permanently, but are **slow** memory devices. The memories are organized in the computer in a manner to achieve high levels of performance at the minimum cost.

Memory Representation

- Memory unit is the amount of data that can be stored in the storage unit. This storage capacity is expressed in terms of Bytes.
- **Bit (Binary Digit)**: A bit is a single binary digit, i.e., 0 or 1. A bit is the smallest unit of representation of data in a computer.
- **Nibble**: A group of 4 bits is called nibble.
- **Byte**: A group of 8 bits is called byte. A byte is the smallest unit, which can represent a data item or a character.
- **Word**: A group of bytes can be further combined to form a word. A word can be a group of 2, 4 or 8 bytes.

contd.



1 bit = 0 or 1

1 Byte (B) = 8 bits

1 Kilobyte (KB) = 1024 Bytes

1 Megabyte (MB) = 1024KB

1 Gigabyte (GB) = 1024 MB

1 Terabyte (TB) = 1024 GB

Types of Memory

1) Primary Memory

2) Secondary Memory

Primary Memory

- **Primary memory** is the main memory of computer which is used to store the information (programs and data) that the computer is currently using. Primary memory is categorized into two main types:
 - Random Access Memory (RAM), and
 - Read Only Memory (ROM)

Random Access Memory (RAM)

- RAM (Random Access Memory) is the internal memory of the CPU for storing data, program and program result.
- It is read/write memory which stores data until the machine is working.
- Access time in RAM is independent of the address that is, each storage location inside the memory is as easy to reach as other locations and takes the same amount of time.
- Data in the RAM can be accessed randomly but it is very expensive.
- RAM is volatile, i.e. data stored in it is lost when we switch off the computer or if there is a power failure. Hence a backup uninterruptible power system (UPS) is often used with computers.
- RAM is small, both in terms of its physical size and in the amount of data it can hold.

Types of RAM

1. Static RAM (SRAM)

- The word static indicates that the memory retains its contents as long as power is being supplied. However, data is lost when the power gets down due to volatile nature.
- SRAM need not have to be refreshed on a regular basis in order to maintain the data. SRAM is used as cache memory and has very fast access. SRAM uses multiple transistors (four to six), for each memory cell. It does not have a capacitor in each cell.

Characteristics of SRAM

- It has long life
- There is no need to refresh
- Faster
- Used as cache memory
- Large size
- Expensive
- Less power consumption

Types of RAM

2. Dynamic RAM (DRAM)

- DRAM, unlike SRAM, must be continually refreshed (recharged) in order to maintain the data. This is done by placing the memory on a refresh circuit that rewrites the data several hundred times per second.
- DRAM is used for most system memory because it is cheap and small. It uses transistors and capacitors. The transistors are arranged in a matrix of rows and columns. The capacitor holds the bit of information 0 and 1.
- The transistor and capacitor are paired to make a memory cell. The transistor acts as a switch that lets the control circuitry on the memory chip read the capacitor or change its state.

Characteristics of DRAM

- It has short data lifetime
- Need to be refreshed continuously
- Slower as compared to SRAM
- Used as RAM
- Lesser in size
- Less expensive
- High power consumption

Read Only Memory (ROM)

- ROM stands for Read Only Memory. The memory from which we can **only read** but **cannot write** on it. This type of memory is **non-volatile**. The information is stored **permanently** in such memories during manufacture.
- A ROM, stores such instructions that are required to start a computer. This operation is referred to as bootstrap. ROM chips are not only used in the computer but also in other electronic items like washing machine and microwave oven.

Types of ROM

1. **MROM (Masked ROM):** The very first ROMs were hard-wired devices that contained a pre-programmed set of data or instructions. These kind of ROMs are known as masked ROMs which are inexpensive.
2. **PROM (Programmable Read only Memory):** PROM is read-only memory that can be modified only once by a user. It can be programmed only once and is not erasable.

Contd.

3. **EPROM (Erasable and Programmable Read Only Memory):** The EPROM can be erased by exposing it to ultraviolet light for a duration of up to 40 minutes.
4. **EEPROM (Electrically Erasable and Programmable Read Only Memory):** The EEPROM is programmed and erased electrically. It can be erased and reprogrammed about ten thousand times.

RAM vs ROM

RAM	ROM
Temporary Storage	Permanent Storage
Store datas in MBs	Store data in GBs
Volatile	Non-volatile
Used in normal operations	Used for startup process of computer

Secondary Memory

- Secondary memory is where programs and data are kept on a long-term basis.
- Common secondary storage devices are the hard disk and floppy disks.
- Secondary memory (or secondary storage) is the **slowest** and **cheapest** form of memory. It **cannot be processed directly** by the CPU. It must first be copied into primary storage (also known as RAM).

Types of Secondary Memory

- Magnetic Tape
- Magnetic Disk
- Optical Disk
 - CD-ROM (Compact Disk - Read Only Memory)
 - CD-R (Compact Disk - Recordable)
 - CD-RW (Compact Disk - Rewritable)
 - DVD-ROM (Digital Versatile Disk - Read Only Memory)
 - DVD-RAM (Digital Versatile Disk - Random Access Memory)

Magnetic Tape

- **Magnetic tape** is the most popular storage medium for large data, which are **sequentially accessed** and **processed**.
- This type of device is still used for large data backup because these are **cheaper** than other devices. Like audio or videotape, the magnetic tape used in computer system which can also be erased and reused indefinitely. Old data on a tape are automatically erased, as new data are recorded in the same area.

Magnetic Disk

- Magnetic disk is a circular disk. The disk rotates with very high speed inside the computer drive.
- Magnetic disks are most popular for **direct access** storage device. Each disk consists of a number of indivisible concentric circles called tracks. Information recorded on tracks of a disk surface in the form of magnetic spots.

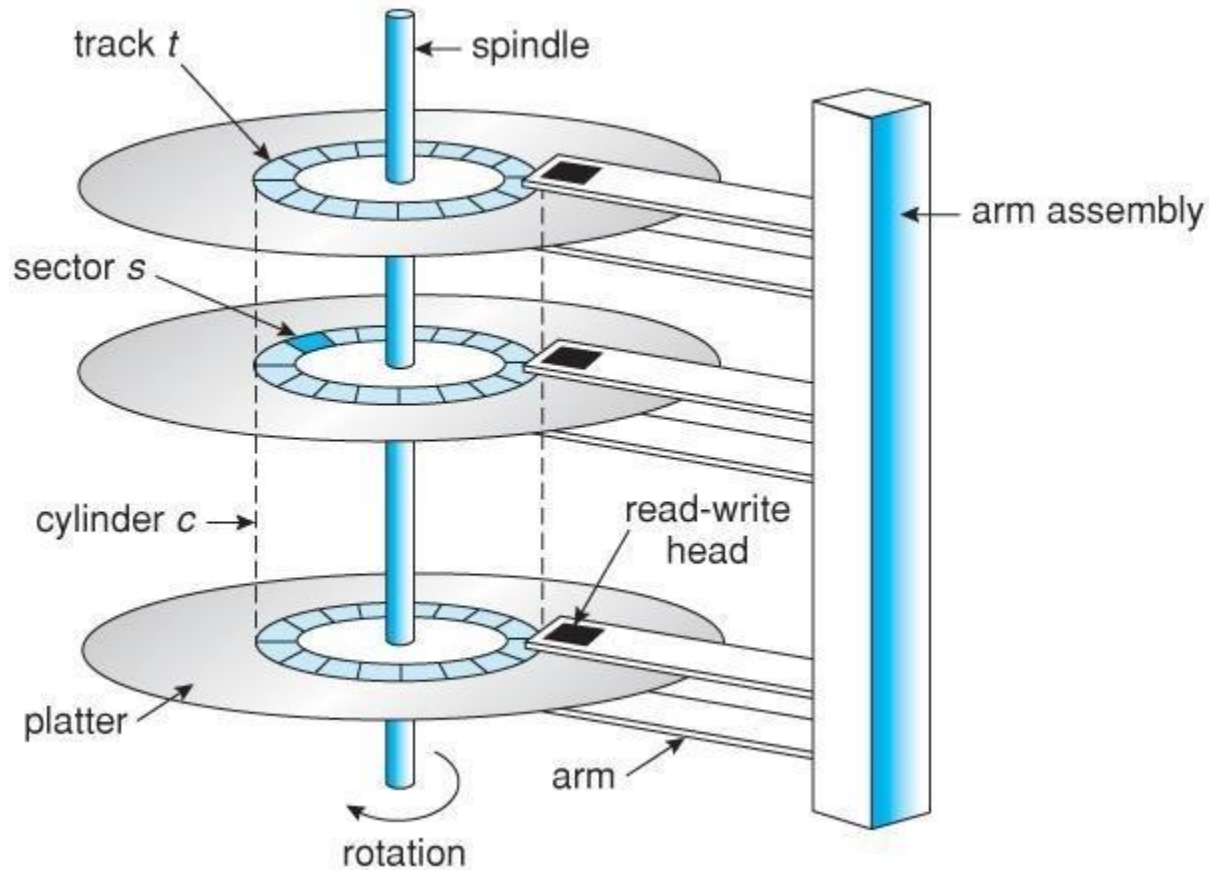
Magnetic Tape vs Magnetic Disk

Magnetic Tape	Magnetic Disk
The cost of magnetic tape is less.	The cost of magnetic disk is high.
Access time for magnetic tape is more.	Access time for magnetic disk is less.
Data transfer rate for magnetic tape is comparatively less.	Data transfer rate for magnetic disk is more.
In magnetic tape data accessing rate is slow.	In magnetic disk data accessing rate is high or fast.

Hard Disk

- The hard disk is direct-access storage medium with a rigid magnetic disk.
- The disk is divided into number of tracks and each tracks are divided into sectors. A hard disk uses round, flat disks called platters, coated on both sides with a special media material designed to store information in the form of magnetic patterns. The platters are mounted by cutting a hole in the center and stacking them onto a spindle. The platters are rotated at very high speed, driven by a special spindle motor connected to the spindle. Each platters consists of READ/WRITE head to read and write data. The disks are rotated at very high speed (usually around 7200 RPM-revolution per minute.)

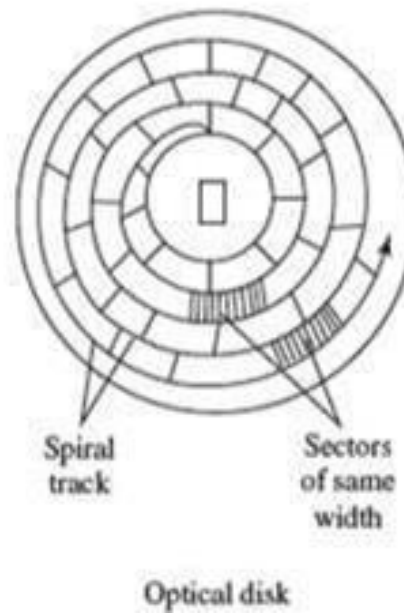
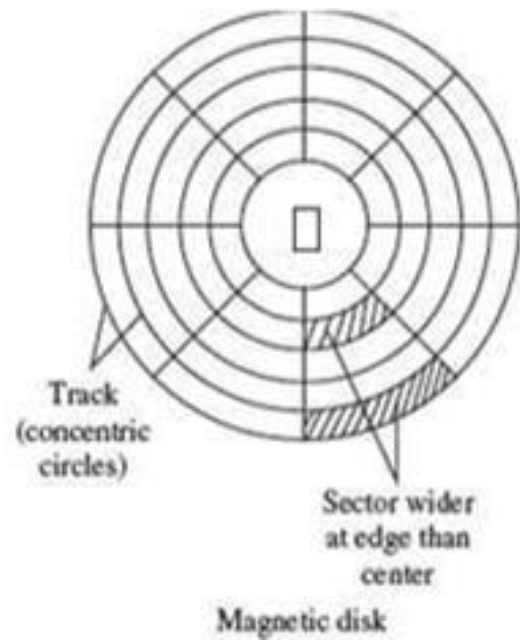
Hard Disk



Optical Disk

- An optical disk is random access storage medium. It is made of glass. Optical disks uses light technology where laser beam is projected and reflected light is observed.
- An optical disk storage system consists of a rotating disk which is coated with a thin metal or some material that is highly reflective. This type of disk uses laser beam technology for READ/WRITE data. Due to the use of laser beam technology, optical disks are also known as laser disk or optical laser disks.
- Unlike magnetic disks, which have several concentric tracks, an optical disk has one long track, which starts at the outer edge and spirals inwards to the center.

Hard Disk



Types of Optical Disks

- 1. CD-ROM (Compact Disk - Read Only Memory):** The data is written onto the CDRom disk before it is sold and cannot be changed by the user. CD-ROMs are used for applications such as distributing software, digital videos or multimedia products.
- 2. CD-R (Compact Disk - Recordable):** A CD-R disk is blank when it is supplied. The user can write data to it just once. After data has been written to the disk it cannot be changed. CD-Rs are often used for making permanent backups of data and distributing software when only a small number of copies are required.

Types of Optical Disks

3. **CD-RW (Compact Disk - Rewritable)**: CD-RW disks can be read from and written to.
4. **DVD-ROM (Digital Versatile Disk - Read Only Memory)**: DVD disks are able to store much more data than CD disks. DVD-ROM disks can be read from but cannot be written to.
5. **DVD-RAM (Digital Versatile Disk - Random Access Memory)**: DVD-RAM disks have all of the benefits of DVD-ROM disks and can be written to as well. These very high capacity disks are ideal for producing backups. Because of their high capacity, DVD disks are used to store high quality video such as complete movies.

Input and Output Devices

- A computer interacts with the external environment via the input-output (I/O) devices.
- Input device is used for providing data and instructions to the computer. After processing the input data, computer provides output to the user via the output device.
- The I/O devices that are attached, externally, to the computer machine are also called peripheral devices.

Input Devices

- The devices which allow users and other applications to input the data and the programs into the computer , for processing are known as "Input Devices".
- Input device can read data and convert them to a form that a computer can use.

Keyboard

- It is a text based input device that allows the user to input alphabets, numbers and other characters. It consists of a set of keys mounted on a board.
- **Alphanumeric keys:** Letters, and numbers **Punctuation keys:** comma, period, semicolon and so on.
Special keys: function keys, control keys, arrows keys cap lock key, and so on.
- For example the original PC keyboard has 84 keys, AT keyboard has also 84 keys, similarly the enhanced keyboard has 101 keys, other keyboard consists up to 104 keys

Mouse

- Mouse is the most popular pointing device. It is a very famous cursor-control device having a small palm size box with a round ball at its base, which senses the movement of the mouse and sends corresponding signals to the CPU when the mouse buttons are pressed.

Mouse Actions

- **Left Click:** Used to select an item.
- **Double Click:** Used to start a program or open a file.
- **Right Click:** Usually used to display a set of commands.
- **Drag and Drop:** It allows you to select and move an item from one location to another. To achieve this place the cursor over an item on the screen, click the left mouse button and while holding the button down move the cursor to where you want to place the item, and then release it.

Trackball

- Track ball is similar to mouse except that it requires less space than mouse. When space is limited trackball is used.
- The user moves the ball directly, while the device itself remains stationary. The user spins the ball in various directions to effect the screen movements. There are usually one to three buttons on trackball.
- Trackball is popular than mouse because it is stationary so it does not require much space and we can place it into any type of surface.

Joystick

- The joystick is a vertical stick which moves the graphic cursor in a direction the stick is moved. It typically has a button on top that is used to select the option pointed by the cursor.
- Joystick is used as an input device primarily used with video games, training simulators and controlling robots.

Digitizing Tablet

- It is an input device used primarily to input drawings, sketches, etc.
- Digitizing tablet is used for Computer Aided Design (CAD) for the design of buildings, automotive designs, and designing of maps, etc.
- Digitizing tablet consists of two parts: electronic tablet and pen. The electronic tablet is a flatbed tablet. The pen looks like a ball pen but has an electronic head.
- The pen is moved on the tablet. Each position on the tablet corresponds to a fixed position on the screen. Drawings can be made on the tablet using a pen, and is provided as input to computer, where, a location on the tablet corresponds to a specific location on the screen.

Light Pen

- It is a pen shaped device used to select objects on a display screen. It is quite like the mouse (in its functionality) but uses a light pen to move the pointer and select any object on the screen by pointing to the object.
- It uses a light-sensitive detector to select objects on display screen.
- It is used for making drawing, graphics and for menu selection. Figures and drawings can be made by moving the pen on computer screen.

Touch Screen

- It allows the user to operate/make selections by simply touching the display screen.
- Common examples of touch screen include information kiosks, and bank ATMs.

Speech recognition Input Device

- The “Microphones - Speech Recognition” is a speech Input device.
- To operate it we require using a microphone to talk to the computer. Also we need to add a sound card to the computer.
- A sound card translates analog audio signals from microphone into digital codes that the computer can store and process. Sound card also translates back the digital sound into analog signals that can be sent to the speakers. A speech recognition program can process the input and convert it into machine-recognized commands or input.

Scanner

- Scanner is an input device, which works more like a photocopy machine.
- It is used when some information is available on paper and it is to be transferred to the hard disk of the computer for further manipulation.
- Scanner is used to input data directly into the computer from the source document without copying and typing the data. The input data to be scanned can be a picture, a text or a mark on a paper.

Optical Character Recognition

- OCR systems are made up of a combination of hardware and software that is used to convert physical documents into machine-readable text.
- Hardware, such as an optical scanner or specialized circuit board is used to copy or read text while software typically handles the advanced processing.
- It is most commonly used to turn hard copy legal or historic documents into PDFs. Once it is done, users can edit, format and search the document as if it was created with a word processor.

Optical Mark Recognition

- OMR is used to detect marks on a paper. The marks are recognized by their darkness. OMR uses an optical mark reader to read the marks.
- The OMR reader scans the forms, detects the mark and passes this information to the computer for processing by application software.
- The optical mark reader detects the presence of mark by measuring the reflected light. OMR is widely used to read answers of objective type tests, where the student marks an answer by darkening a particular circle using a pencil. OMR is also used to read forms, questionnaires, order forms, surveys, etc.

Magnetic Ink Character Recognition

- MICR is a technology used to verify the legitimacy or originality of paper documents, especially checks.
- Special ink, which is sensitive to magnetic fields, is used in the printing of certain characters on the original documents. Information can be encoded in the magnetic characters.
- The use of MICR can enhance security and minimize the losses caused by some types of crime. If a document has been forged - for example, a counterfeit check is produced, the magnetic-ink line will either not respond to magnetic fields, or will produce an incorrect code when scanned using a device designed to recover the information in the magnetic characters.

Barcode Reader

- Barcodes are adjacent vertical lines of different width that are machine readable.
- Goods available at supermarkets, books, etc. use barcode for identification. Barcodes are read using reflective light by barcode readers. This information is input to the computer which interprets the code using the spacing and thickness of bars.
- Hand-held barcode readers are generally used in departmental stores to read the labels, and in libraries to read labels on books. Barcode readers are fast and accurate.

Output Devices

- Output devices provide output to the user, which is generated after processing the input data. The processed data, presented to the user via the output devices could be text, graphics, audio or video.
- The output could be on a paper or on a film in a tangible form, or, in an intangible form as audio, video and electronic form.

Printer

- A printer is an external hardware output device that takes the electronic data stored on a computer or other device and generates a hard copy.
- Printers are one of the most popular computer peripherals and are commonly used to print text and photos.

Plotter

- Plotters are used to print graphical output on paper. It interprets computer commands and makes line drawings on paper using multicolored automated pens. It is capable of producing graphs, drawings, charts, maps etc.
- Computer Aided Engineering (CAE) applications like CAD (Computer Aided Design) and CAM (Computer Aided Manufacturing) are typical usage areas for plotters.

Monitor

- A monitor is an electronic visual computer display.
- The monitor is associated with a keyboard for manual input of characters and displays the information as it is keyed in. It also displays the program or application output.
- Older computer monitors made use of cathode ray tubes (CRT), which made them large, heavy and inefficient. Nowadays, flat-screen LCD monitors are used in devices like laptops, PDAs and desktop computers because they are lighter and more energy efficient.

Audio and Video Output

- Screen image projector or data projector is an output device that displays information from the computer onto a large white screen.
- The Audio output is the ability of the computer to output sound. Audio output device are speakers, headset or headphone.

Interfaces

- **Software Interface:** When referring to software, an interface is a program that allows a user to interact computers in person or over a network. An interface may also refer to controls used in a program that allow the user to interact with the program. One of the best examples of an interface is a GUI (Graphical User Interface). This type of interface is how you navigate your computer.
- **Hardware Interface:** When referring to hardware, an interface is a physical device, port, or connection that interacts with the computer or other hardware device. For example, Hardware interfaces refer to the connection and communication of different devices. For example, a printer being connected to a USB port is actually connected to the USB interface of the system. Another example of a hardware interface is a graphics card plugged into a PCI slot, which is using the PCI interface to connect and communicate.

Serial Port vs Parallel Port

Serial Port	Parallel Port
Serial Port is used for serial data transmission.	Parallel Port is used for parallel data transmission.
Transmission speed of a serial port is slow as compared to a parallel port.	Transmission speed of a parallel port is quite high as compared to a serial port.
Wire connections to serial port are quiet less as compared to parallel port.	No. of wires that are connected to parallel port are quiet high as compared to serial port.
A serial port is able to transmit a single stream of data at a time.	A parallel port is able to transmit multiple data streams at a time.
A serial port sends data bit by bit after sending a bit at a time.	A parallel port sends data by sending multiple bits in parallel fashion.
Modems, security cameras, device controllers use serial ports.	Printers, Hard Drives, CD drives use parallel ports.